

» NOTABLE EXPLORERS (CONTINUED)

NAME	LIVED	ORIGIN	EXPEDITIONS/DISCOVERIES
Neil Armstrong	b. 1930	US	Apollo 11 (July 20, 1969); first man to walk on the Moon
Vladimír Remek	b. 1948	Czechoslovakia	Soyuz 28 (March 2, 1978); first person in space from a country other than the US or USSR
Sigmund Jähn	b. 1937	German Democratic Republic	Soyuz 31 (August 26, 1978); first German in space
Jean-Loup Chrétien	b. 1938	France	Soyuz T-11 (June 24, 1982); first French person in space
Ulf Merbold	b. 1941	Germany	STS-9 28 (November 1983); first ESA astronaut, second German in space
Rakesh Sharma	b. 1949	India	Soyuz T-11 (April 3, 1984); first Indian in space
Sultan Salman al Saud	b. 1956	Saudi Arabia	STS-56 (June 17, 1985); first Arab (and first Muslim) in space
Mamoru Mori	b. 1948	Japan	STS-47 (September 12, 1992); first Japanese person in space
Valeri Polyakov	b. 1942	Russia	Soyuz TM-18 (January 8, 1998); longest space flight at 437 days
John Glenn	b. 1921	US	STS-95 (October 29, 1998); oldest person in space at 77 years old—previously flew on Friendship 7 in 1962
Dennis Tito	b. 1940	US	Soyuz TM 32 (April 28, 2001); first “space tourist”
Yang Liwei	b. 1965	China	Shenzhou 5 (October 15, 2003); first Chinese person in space
Sergei Krikalev	b. 1958	Russia	Soyuz TMA-6 (October 11, 2005); reached most time spent in space (803 days 9 hours 39 minutes)

INVENTIONS AND DISCOVERIES

The modern world is very different from the world of our ancestors. Over the course of human existence, basic human needs—from the need to survive to the urge to obtain knowledge—have produced tens of thousands of inventions and discoveries. These have transformed both the way we function and the way we think, and have made us distinct from the rest of the animal kingdom.

» NOTABLE INVENTIONS AND DISCOVERIES

INVENTION/DISCOVERY	DATE	ORIGINATOR	PLACE OF ORIGIN
Stone tools	c. 2.75 MYA	Early humans	Africa
Control of fire	c. 400,000 YA	Early humans	Africa
Boat	c. 50,000 YA	Early migrants	Australasia
Mining	c. 40,000 YA	Paleolithic humans	Europe
Permanent shelters	c. 28,000 YA	Paleolithic humans	Eastern Europe
Pottery vessels	c. 14,000 BCE	Jomon people	Ancient Japan
Farming	c. 10,000 BCE	West Asian peoples	Mesopotamia
Irrigation	c. 5000 BCE	West Asian peoples	Mesopotamia
Horse domestication	c. 4500 BCE	Andronovo culture	Europe/Asia
Plough	c. 4000 BCE	Sumerian people	Mesopotamia
Wheeled transport	c. 3500 BCE	Sumerian people	Sumer
Silk weaving	c. 3500 BCE	Chinese peoples	Ancient China
Writing	c. 3300 BCE	Sumerian people	Sumer/Egypt
Calendar	c. 3000 BCE	Babylonians	Babylonia

INVENTION/DISCOVERY	DATE	ORIGINATOR	PLACE OF ORIGIN
Papyrus scroll	c. 2600 BCE	Imhotep (attributed)	Ancient Egypt
Plumbing	c. 2500 BCE	Indus Valley civilization	Indus (Pakistan)
Law code	c. 1755 BCE	King Hammurabi	Babylonia
Alphabet	14th century BCE	Semitic people (slaves of the Egyptians)	Ancient Egypt
Magnetism	c. 1000 BCE	Thales of Miletus (attributed)	Ancient Greece
Coinage	c. 600 BCE	Lydian people	Ancient Turkey
World map	6th century BCE	Babylonians	Babylonia
Planetary models	c. 360 BCE	Eudoxus of Cnidus	Ancient Greece
Rotation of the Earth	c. 350 BCE	Heraclides Ponticus	Ancient Greece
Steel production	c. 200 BCE	Han dynasty	India/China
Compound pulley	c. 200 BCE	Archimedes	Ancient Greece
Encyclopaedia	77	Pliny the Elder	Roman Empire
Paper	c. 105	Cai Lun	Ancient China
Compass	250	Chinese peoples	Ancient China
Concept of zero/decimal system	c. 590–650	Brahmagupta	India
Astrolabe	c. 800	Muhammad al-Fazari	Arabia
University	859	Fatimah al-Fihri	Morocco
Star chart	c. 1000–50	Abu Rayhan Biruni	Persia
Pendulum	c. 000	Ibn Yunus	Egypt
Magnifying glass	c. 1021	Ibn al-Haytham	Persia
Moveable type	c. 1045	Bi Sheng	China
Mechanical clock	1088	Su Song	China
Algebra	1202	Fibonacci	Italy
The scientific method	c. 1220–35	Robert Grosseteste	England
Printing press	c. 1445	Johannes Gutenberg	Germany
Terrestrial globe	c. 1490	Martin Behaim	Bohemia
Sun-centred Universe	1503–43	Copernicus	Italy
Compound microscope	c. 1595	Hans Lippershey, Zacharias Janssen	Netherlands
Laws of planetary motion	1609–19	Johannes Kepler	Germany
Newspaper	1609	Johann Carolus	Germany
Refracting telescope	1609	Galileo Galilei	Italy
Mechanical calculator	1642	Wilhelm Schickard	Germany
Barometer	1643	Evangelista Torricelli	Italy
Atmospheric pressure	1647–48	Blaise Pascal	France
Microscopic life	1673	Antoni van Leeuwenhoek	Netherlands
Laws of motion	1687	Sir Isaac Newton	England
Seed drill	1701	Jethro Tull	England
Steam piston engine	1712	Thomas Newcomen	Britain
Marine chronometer	1735	John Harrison	Britain
Lightning rod	1752	Benjamin Franklin	US
Watt steam engine	1776	James Watt	Britain
Oxygen	1777	Antoine Lavoisier	France
Hot air balloon	1783	Montgolfier brothers	France
Threshing machine	1786	Andrew Meikle	Britain
Battery	1800	Alessandro Volta	Italy
Bicycle	1818	Karl Drais	Germany
Permanent photography	c.1820	Joseph Nicéphore Niépce	France
Braille alphabet	1821	Louis Braille	France
Electric motor	1821	Michael Faraday	Britain

